

DreamCampus — Physics PYQ Practice

Original exam-style objective questions with answers and revision guidance

About this set

These are original PYQ-pattern questions, not verbatim reproductions of copyrighted past papers. They are designed for objective-exam practice across the common Physics concepts used by WBJEE/JEE Main/NEET. Check official exam documents for the current syllabus and pattern.

1. Kinematics

A particle has initial velocity 8 m/s and acceleration 3 m/s² for 4 s. What is its final velocity?

- A) 12 m/s
- B) 20 m/s
- C) 24 m/s
- D) 32 m/s

Answer: B

2. Kinematics

A body starts from rest with acceleration 2 m/s². Distance travelled in 5 s is:

- A) 10 m
- B) 20 m
- C) 25 m
- D) 50 m

Answer: C

3. Newton's Laws

A 4 kg body has a net force of 12 N. Its acceleration is:

- A) 2 m/s²
- B) 3 m/s²
- C) 4 m/s²
- D) 48 m/s²

Answer: B

4. Friction

For a block on a horizontal surface, limiting friction in the simple model is proportional to:

- A) area only
- B) normal reaction
- C) velocity only
- D) time only

Answer: B

5. Work Energy

A 2 kg body moving at 5 m/s has kinetic energy:

- A) 5 J

B) 10 J

C) 20 J

D) 25 J

Answer: C

6. Circular Motion

If speed is doubled at the same radius, centripetal force becomes:

A) half

B) double

C) four times

D) unchanged

Answer: C

7. Gravitation

If separation between two point masses becomes 3 times, gravitational force becomes:

A) $F/3$

B) $F/6$

C) $F/9$

D) $9F$

Answer: C

8. Fluids

For steady incompressible flow, the product of cross-sectional area and flow speed is:

A) variable arbitrarily

B) constant

C) always zero

D) equal to pressure

Answer: B

9. Thermodynamics

For an ideal gas undergoing an isothermal process, temperature is:

A) increasing

B) decreasing

C) constant

D) zero

Answer: C

10. SHM

For ideal SHM, acceleration is:

A) proportional to $+x$

B) proportional to $-x$

C) constant

D) independent of x

Answer: B

11. Waves

If frequency is doubled while wave speed remains constant, wavelength becomes:

- A) doubled
- B) halved
- C) unchanged
- D) four times

Answer: B

12. Electrostatics

Electric potential at a point due to a positive point charge is:

- A) negative
- B) positive in the usual zero-at-infinity convention
- C) always zero
- D) vector

Answer: B

13. Capacitors

For a capacitor, capacitance is defined by:

- A) $C=V/Q$
- B) $C=Q/V$
- C) $C=QV$
- D) $C=Q+V$

Answer: B

14. Current

A $6\ \Omega$ resistor is connected to 12 V. Current is:

- A) 0.5 A
- B) 2 A
- C) 6 A
- D) 72 A

Answer: B

15. Magnetism

Magnetic force on a charge moving exactly parallel to $\mathbf{B}$ is:

- A) maximum
- B) zero
- C) qB
- D) q/B

Answer: B

16. EMI

The negative sign in Faraday's law represents:

- A) charge conservation

- B) Lenz's law
- C) Ohm's law
- D) Coulomb's law

Answer: B

17. Optics

The relation $n \sin i = n \sin r$ is:

- A) Newton's law
- B) Snell's law
- C) Ampere's law
- D) Gauss's law

Answer: B

18. Modern Physics

Photon energy is proportional to:

- A) wavelength
- B) frequency
- C) mass only
- D) radius

Answer: B

19. Semiconductors

A p-n junction diode conducts readily in:

- A) forward bias
- B) reverse bias only
- C) zero field only
- D) vacuum

Answer: A

20. Mixed

Which quantity is scalar?

- A) displacement
- B) velocity
- C) electric field
- D) electric potential

Answer: D

Revision Analysis

After each attempt, mark questions as: C = correct and confident, G = guessed, W = wrong. Re-solve every W and G question without looking at the answer.